1. PEAS Specification: Delivery Robot

PEAS stands for Performance Measure, Environment, Actuators, and Sensors. It describes an intelligent agent's task environment.

Component	Concrete entries
P — Performance Measure	1. Deliver parcels to the correct destination. 2. Minimize delivery time and battery consumption. 3. Avoid collisions and protect parcels.
E — Environment	1. University campus pathways and corridors. 2. Students, pedestrians, doors, elevators, and obstacles. 3. Delivery stations and parcel drop-off locations.
A — Actuators	1. Drive motors and steering system. 2. Robotic arm or parcel compartment lock. 3. Speaker and display for delivery notifications.
S — Sensors	1. Cameras for detecting people and obstacles. 2. LiDAR or ultrasonic sensors for distance measurement. 3. GPS or indoor localization sensors for navigation.

Operational assumptions: The robot operates on a university campus, has a digital campus map, and can access designated delivery locations. It must stop when a route is unsafe and request assistance if it cannot continue.

Performance trade-off: Faster delivery can reduce travel time, but moving too quickly increases collision risk. The robot should prioritize safety over speed.

2. PEAS Specification: LLM-Based Student-Support Agent

Component	Concrete entries
P — Performance Measure	1. Provide accurate answers to student questions. 2. Respond quickly and clearly. 3. Protect student privacy and avoid unsupported information.
E — Environment	1. University learning management system (LMS). 2. University course materials, assignment instructions, and academic policies. 3. Student-support chat interface.
A — Actuators	1. Generate text responses and explanations. 2. Provide links to relevant course resources. 3. Submit a support ticket or escalate a question to a human staff member.
S — Sensors	1. Student messages and questions. 2. Retrieved course documents and university policy information. 3. Tool results, such as assignment deadlines or ticket status.

Operational assumptions and tool access: The agent can access the student-support chat, approved course documents, a read-only LMS search tool, and a support-ticket creation tool. It cannot access private grades, change marks, or modify student records unless explicitly authorized. It should identify uncertainty and refer sensitive or unresolved issues to authorized university staff.

Performance trade-off: Providing an immediate answer is convenient, but answering without checking reliable course materials may produce incorrect information. The agent should prioritize verified answers over speed and escalate questions when evidence is insufficient.

Conclusion

Both agents use the PEAS framework, but their environments and actions differ. The delivery robot interacts with the physical world through motors and sensors, whereas the LLM-based student-support agent interacts with students and approved digital resources. Their performance measures should reflect their main responsibilities: physical safety and reliable delivery for the robot, and accuracy, privacy, and useful academic support for the LLM agent.